

PicoCTF Write-Up

Cooking solutions, one flag at a time.

General Skills & Cryptography | Beginner | June 2026

■ Introduction

Five challenges solved in one session. This write-up covers number bases (hex, binary, Base64), ROT13 cryptography, and Linux help flags — all broken down from scratch so anyone starting from zero can follow along.

GENERAL SKILLS

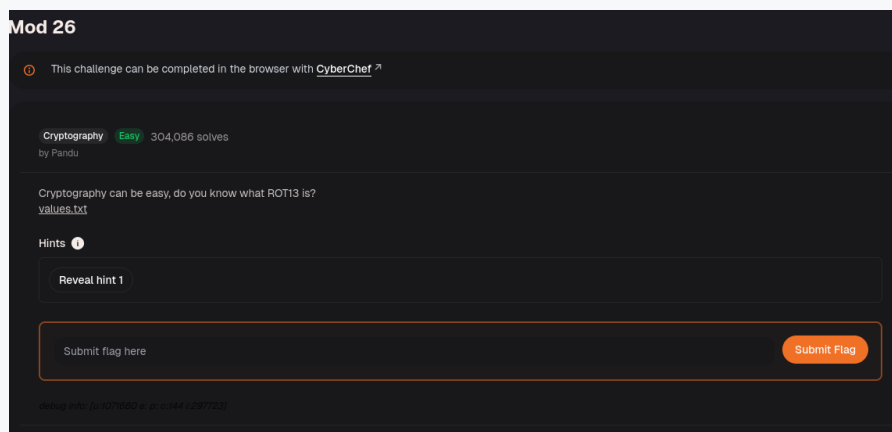
CRYPTOGRAPHY

EASY

BEGINNER

■ Challenge 1 — Mod 26 (ROT13)

■ PicoCTF — Mod 26 Challenge Page



What's the challenge asking?

Decode a ROT13-encoded string to find the flag. The challenge gives you a **values.txt** file with the encoded text.

Concept: What is ROT13?

ROT13 stands for **Rotate by 13**. Every letter shifts 13 places forward in the alphabet. Since the alphabet has 26 letters, applying ROT13 twice brings you back to the original — the same operation both encodes and decodes. It's not real encryption, just obfuscation. In CTFs it's an intro to **substitution ciphers**.

Encoded text:

```
cvpbPGS{arkg_gvzr_V'yy_ge1_2_ebhaqf_bs_ebg13_45559noq}
```

Command Used

```
$ cat values.txt | tr 'A-Za-z' 'N-ZA-Mn-za-m'
```

```
tr      -> translate/replace characters
A-Za-z  -> all letters (input)
N-ZA-Mn-za-m -> ROT13 shifted alphabet (output)
```

■ Flag Captured

picoCTF{next_time_I'll_try_2_rounds_of_rot13_4559abd}

Key Takeaways

- ROT13 = substitution cipher. Every letter shifts 13 places.
- Encoding ≠ Encryption. ROT13 has no key — anyone can reverse it.
- The **tr** command translates characters in Linux — useful for simple ciphers.

■ Challenge 2 — Warmed Up (Hex → Decimal)

■ PicoCTF — Warmed Up Challenge Page

General Skills **Easy** 161,946 solves
by Sanjay C/Danny Tunitis

What is 0x3D (base 16) in decimal (base 10)?

Hints ⓘ

Reveal hint 1

Submit flag here Submit Flag

debug info: [u:1076600 a:pr:55 (287405)]

What's the challenge asking?

Convert **0x3D** from hexadecimal (base 16) to decimal (base 10).

Concept: Hexadecimal

Hex uses 16 digits: 0–9 and A–F, where A=10, B=11, C=12, D=13, E=14, F=15. The **0x** prefix just tells you the number is in hex. You'll see hex everywhere in cybersecurity — memory addresses, color codes, byte values.

Working it out:

0x3D -> 3D in hex

```
3 x 16 = 48
D = 13
48 + 13 = 61
```

■ Flag Captured

picoCTF{61}

■ Challenge 3 — 2warm (Decimal → Binary)

■ PicoCTF — 2warm Challenge Page

Can you convert the number 42 (base 10) to binary (base 2)?

Hints ⓘ

Reveal hint 1

Submit flag here Submit Flag

debug info: {u:1071660 e:p:c86 l:297066}

What's the challenge asking?

Convert **42** from decimal (base 10) to binary (base 2).

Concept: Binary

Binary uses only two digits: 0 and 1. Every piece of data in a computer is ultimately stored as binary. To convert decimal to binary, keep dividing by 2 and read the remainders from bottom to top.

Working it out:

```
42 / 2 = 21 remainder 0
21 / 2 = 10 remainder 1
10 / 2 = 5 remainder 0
5 / 2 = 2 remainder 1
2 / 2 = 1 remainder 0
1 / 2 = 0 remainder 1
```

Read remainders bottom to top: **101010**

■ Flag Captured

picoCTF{101010}

■ Challenge 4 — Bases (Base64)

What does this bDNhcm5fdGgzX3lwcDM1 mean? I think it has something to do with bases.

Hints ⓘ

Submit your answer in our flag format. For example, if your answer was 'hello', you would submit 'picoCTF{hello}' as the flag.

What's the challenge asking?

Decode the string **bDNhcm5fdGgzX3lwcDM1** — it has something to do with bases.

Concept: Base64

Base64 is a way of encoding binary data into readable text using 64 characters (A–Z, a–z, 0–9, +, /). It's not encryption — anyone can decode it instantly. You'll see it in web tokens, email attachments, and CTFs constantly.

Command Used

```
$ echo "bDNhcm5fdGgzX3lwcDM1" | base64 -d
```

```
echo      -> prints the string
|         -> pipes output into the next command
base64 -d -> decode from Base64 (-d = decode)
```

■ Terminal — Base64 decode result

```
(kali㉿kali)-[~]
-$ echo "bDNhcm5fdGgzX3lwcDM1" | base64 -d
l3arn_th3_r0p35

(kali㉿kali)-[~]
-$
```

■ Flag Captured

picoCTF{l3arn_th3_r0p35}

■ Challenge 5 — Warm (Help Flags)

■ PicoCTF — Warm Challenge Hints

Can you invoke help flags for a tool or binary? This program has extraordinarily helpful information...
warm

Hints ⓘ

This program will only work in the webshell or another Linux computer.

To get the file accessible in your shell, enter the following in the Terminal prompt: `$ wget <URL here>`, where the url can be found in the details section.

Run this program by entering the following in the Terminal prompt: `$ ./warm`, but you'll first have to make it executable with `$ chmod +x warm`

`-h` and `--help` are the most common arguments to give to programs to get more information from them!

Not every program implements help features like `-h` and `--help`.

What's the challenge asking?

Download a binary called **warm**, make it executable, and pass it a help flag to reveal the hidden flag.

Steps

```
# Step 1 — Download the file
$ wget <URL from challenge page>

# Step 2 — Make it executable
$ chmod +x warm

# Step 3 — Run it first to see what it wants
$ ./warm
Hello user! Pass me a -h to learn what I can do!

# Step 4 — Pass the help flag
$ ./warm -h
```

Mistake I Made

```
X $ -h          -> -h: command not found
X $ -- help     -> --: command not found

-h is not a standalone command.
It must be passed TO the program: ./warm -h
```

Terminal — wget, chmod +x, ./warm -h, flag captured

```
(kali@kali)-[~]
└─$ wget https://challenge-files.picoctf.net/c_wily_courier/1e14db3a752e16eae2b0e0d73d9779f9c4ddfd8942f60f3285a2986068480316/warm
--2026-06-03 12:47:03-- https://challenge-files.picoctf.net/c_wily_courier/1e14db3a752e16eae2b0e0d73d9779f9c4ddfd8942f60f3285a2986068480316/warm
Resolving challenge-files.picoctf.net (challenge-files.picoctf.net)... 108.157.78.26, 108.157.78.33, 108.157.78.84, ...
Connecting to challenge-files.picoctf.net (challenge-files.picoctf.net)|108.157.78.26|:443 ... failed: Connection refused.
Connecting to challenge-files.picoctf.net (challenge-files.picoctf.net)|108.157.78.33|:443 ... failed: Connection refused.
Connecting to challenge-files.picoctf.net (challenge-files.picoctf.net)|108.157.78.84|:443 ... connected.
HTTP request sent, awaiting response... 200 OK
Length: 19312 (19K) [application/octet-stream]
Saving to: 'warm'

warm                               100%[=====] 18.86K
2026-06-03 12:47:48 (11.0 MB/s) - 'warm' saved [19312/19312]

(kali@kali)-[~]
└─$ chmod +x warm

(kali@kali)-[~]
└─$ ./warm
Hello user! Pass me a -h to learn what I can do!

(kali@kali)-[~]
└─$ -h
-h: command not found

(kali@kali)-[~]
└─$ --help
--: command not found

(kali@kali)-[~]
└─$ ./warm -h
Oh, help? I actually don't do much, but I do have this flag here: picoCTF{b1scu1ts_4nd_gr4vy_ac5832c}
```

Flag Captured

picoCTF{b1scu1ts_4nd_gr4vy_ac5832c}

Key Takeaways

- **wget** downloads files from the internet directly in the terminal.
- **chmod +x** makes a file executable — without it the OS won't run it.
- **./** means run this file from the current directory.
- **-h** and **--help** are flags you pass TO a program, not standalone commands.
- Always read error messages — they tell you exactly what went wrong.

Concepts Covered This Session

Concept	What it is	Used in
ROT13	Shift every letter by 13 places	Mod 26
Hexadecimal	Base 16 number system (0–9, A–F)	Warmed Up
Binary	Base 2 number system (0 and 1)	2warm
Base64	Encoding binary data as readable text	Bases
chmod +x	Make a file executable in Linux	Warm
wget	Download files from the internet	Warm
-h / --help	Pass help flag to a program	Warm

GitHub

LinkedIn

X

